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To cite this article: Fathima Fasmin and Ramanathan Srinivasan 2017 *J. Electrochem. Soc.* **164** H443

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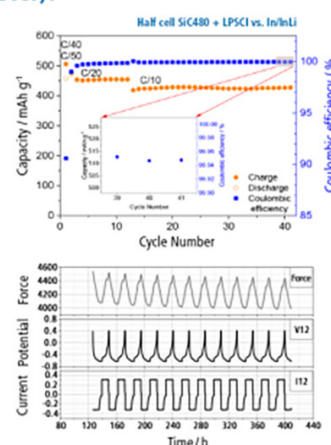
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Review—Nonlinear Electrochemical Impedance Spectroscopy

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Nonlinear Electrochemical Impedance Spectroscopy (NLEIS) is a method of characterizing an electrochemical systems by applying a large amplitude sinusoidal perturbation signal. This article presents a comprehensive review of existing literature on NLEIS analysis development and applications. Two types of NLEIS reports available in the literature, viz. one in which only the fundamental impedance is studied and the other in which both the fundamental and higher harmonic responses are analyzed, are reviewed. Early reports on the development of NLEIS methodology are compiled. Recent advances in the application of NLEIS, in particular in the area of fuel cells, are summarized. In addition, applications of related techniques such as total harmonic distortion and Volterra kernel are included in the purview of this article. There is a large scope for applying NLEIS to understand electrochemical processes. The roadblocks that need to be overcome to enable wider application of NLEIS are identified, and directions for further research are suggested.

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Manuscript submitted March 20, 2017; revised manuscript received April 24, 2017. Published May 6, 2017.

Electrochemical impedance spectroscopy (EIS) is a versatile technique used to characterize phenomena such as corrosion, and systems such as fuel cells and batteries.^{1,2} In the strictest sense, the term “EIS” can be employed only under small signal conditions, i.e. when the system response is linear.³ However, under small signal conditions, the kinetic information present in the nonlinear part of the response would be missing. In addition, small amplitude perturbation often leads to poor signal-to-noise ratio. Therefore, interest in the so called nonlinear EIS (NLEIS) is growing now, and measurements performed with large amplitude perturbations are reported as NLEIS data in literature.^{4–10}

NLEIS method can be thought of as an extended version of EIS technique wherein a large amplitude perturbation is applied to the system and the response is recorded at the fundamental, and at higher harmonics.^{4–10} In many cases, only the fundamental harmonic impedance is measured and its variation with the amplitude of the perturbation signal is regarded as nonlinear effect.^{11–15} Early studies on NLEIS were targeted toward faster estimation of charge transfer coefficient,¹⁶ or more accurate estimation of polarization resistance by extrapolation.^{12,17} Under large signal conditions, the signal to noise ratio is good. On the other hand, the analysis of NLEIS is more challenging than that of corresponding small amplitude EIS results, mainly because the linearized equations are often amenable to analytical solutions while obtaining analytical solutions to the nonlinear equations is difficult.^{4–9,11–15} Nevertheless, in recent years there has been a renewed interest in the measurement and analysis of response to large amplitude perturbations. Fig. 1 shows an overview of the timeline representing the theoretical and experimental development of nonlinear EIS, along with its recent applications in electrochemical research, which include redox reactions, energy storage devices like fuel cells, batteries, biological materials etc. A short review focused on the investigation of solid oxide fuel cell (SOFC) electrode reaction mechanism using nonlinear EIS was published by McDonald and Adler,¹⁸ but its scope was limited to SOFC electrode reactions. To the best of our knowledge, a review article encompassing wider range of research studies on experimental and theoretical aspects of NLEIS is currently not available and this paper is an attempt to fill that gap.

In this paper, we summarize the published reports on NLEIS, in particular for electrochemical reactions and identify a few areas where we believe that further work will be fruitful. Early literature reports present the results at fundamental frequency and its dependence on the input perturbation amplitude,^{11–17,19–23} while recent articles report the results at higher harmonics.^{4–10} Both types of nonlinear EIS reports are summarized here. In addition, a few reports employ related

measures of nonlinearity such as Volterra kernels^{6,7,24} and total harmonic distortion (THD)^{25,26} and they are also included in the scope of this article. This review is organized as follows. For the sake of completeness, traditional EIS technique is introduced and NLEIS is developed as a natural extension of linear EIS. Next, the reports on the development of NLEIS analysis methods, and the experimental validations are described. The studies on application of nonlinear EIS to characterize electrochemical systems are compiled. The challenges to be overcome before NLEIS application becomes more widespread, and the future scope are summarized.

Brief Overview of EIS

Impedance spectroscopy is a technique wherein sinusoidal perturbation at a particular frequency is applied onto a system and the response is recorded. This procedure is repeated over a range of frequencies in order to obtain a spectrum.^{1,3} In most systems, pseudo potentiostatic measurement is common.²⁷ The perturbation can be a sinusoidal potential wave of amplitude E_{ac0} superimposed on a constant dc potential E_{dc} , and the magnitude (i_1) and phase (ψ_1) of the current density are recorded.²⁷ In general, the current response to applied potential $E = E_{dc} + E_{ac0} \sin(\omega t)$ can be written in Fourier series as

$$i = i_0 + i_1 \sin(\omega t + \psi_1) + i_2 \sin(2\omega t + \psi_2) + \dots + i_n \sin(n\omega t + \psi_n) + \dots \quad [1]$$

The vector ratio of potential to current is denoted as impedance. Differential impedance, often called just impedance in electrochemistry community for the sake of brevity, is defined as $Z = \delta E / \delta i$, and it has a magnitude of E_{ac0} / i_1 and a phase of $-\psi_1$. Since it has both magnitude and phase, it is usually represented as a complex quantity, for convenience. In energy related systems such as fuel cells and batteries, measurement in pseudo galvanostatic mode is common.^{28,29} Here, a sinusoidal current wave of amplitude i_{ac0} at an angular frequency ω , superimposed on a dc current i_{dc} is applied and the potential magnitude (E_1) and phase (ξ_1) at the applied frequency are recorded. Under pseudo galvanostatic mode, with an applied current of $i = i_{dc} + i_{ac0} \sin(\omega t)$, the measured potential can be written as

$$E = E_0 + E_1 \sin(\omega t + \xi_1) + E_2 \sin(2\omega t + \xi_2) + \dots + E_n \sin(n\omega t + \xi_n) + \dots \quad [2]$$

The magnitude of the impedance is then E_1 / i_{ac0} , while its phase is ξ_1 . If the perturbation is small enough, then the system response will be linear and the impedance will only be a function of the applied angular frequency (ω). The measured impedance will not change if the perturbation amplitude is reduced, i.e. Z is not a function of E_{ac0} or i_{ac0} , and higher harmonics will be negligible compared to the result at fundamental frequency (i_n / i_1 or E_n / E_1 will be ≈ 0 for all $n > 1$)

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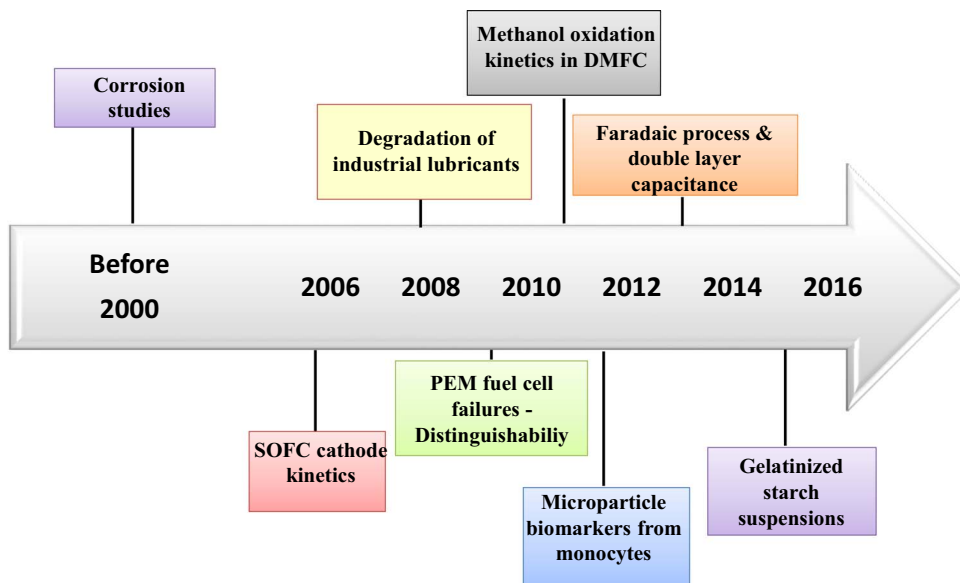


Figure 1. Time line showing a few key research activities based on NLEIS.

and this greatly simplifies the analysis.^{1,3,30} However, electrochemical systems are inherently nonlinear,^{1,3} and under large signal conditions, the linearity assumptions are no longer valid and the nonlinear terms must be accounted for.

During EIS experiments the amplitude of the applied potential perturbation is maintained low as possible (typically ≤ 20 mV), so that the system response can be approximated to be linear while maintaining an acceptable signal to noise ratio. Fig. 2a shows an electrode electrolyte interface, and in case of a reaction with an adsorbed intermediate, the impedance spectra can be represented by an equivalent electrical circuit given in Fig. 2b.³¹ The spectrum can be plotted in various forms,¹ and often it is shown either as a complex plane plot (Fig. 2c) or as Bode plot (Fig. 2d). Since impedance is represented as

a complex quantity, it can be written as $Z = Z_{Re} + j Z_{Im}$, where $j = \sqrt{-1}$. In the complex plane plot presented in Fig. 2c, the frequencies are not shown but usually a few frequency points are marked. Two semi-circles are present in the complex plane plot and the semi-circle at high frequencies arises from the double layer capacitance of the electrode-electrolyte interface and the charge transfer resistance (R_t), while the semi-circle at low frequencies arises from the faradaic reaction. The high frequency limit of the total impedance corresponds to solution resistance (R_{sol}), while the low frequency limit of the total impedance (Z_{LF}) corresponds to the sum of polarization resistance (R_p) and R_{sol} , as shown in Fig. 2c. In addition, if the high frequency semi-circle is extended to the real axis, the intercept would correspond to the sum of R_t and R_{sol} , as marked in Fig.

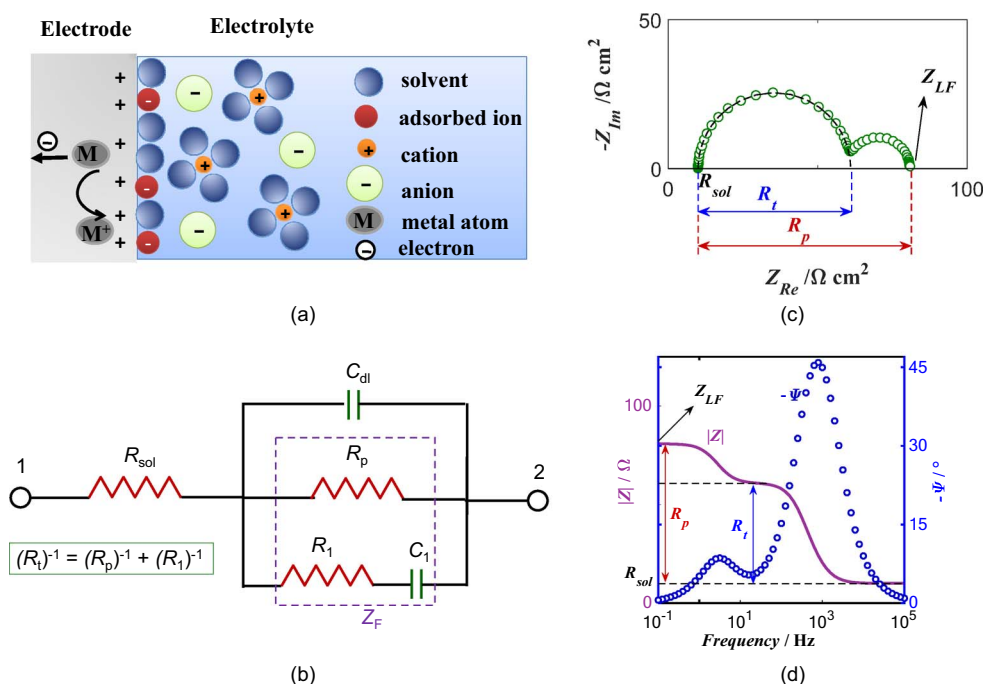


Figure 2. (a) Cartoon of double layer structure, (b) Equivalent electrical circuit (EEC) corresponding to a two-step reaction under small signal conditions. Impedance spectrum as (c) complex plane plot and as (d) Bode Plot.

2c. These limiting values are also marked in the Bode plot (Fig. 2d). The charge transfer resistance is the impedance of the faradaic process (Z_F) at the high frequency limit.

While solution resistance is always assumed to be independent of perturbation amplitude, *sensu stricto*, R_t and R_p are defined at vanishingly small perturbation amplitudes as given in Eqs. 3b and 3c^{13–15,22}

$$R_{sol} = \lim_{\omega \rightarrow \infty} Z_T \quad [3a]$$

$$R_t = \lim_{E_{ac0} \rightarrow 0} \lim_{\omega \rightarrow \infty} Z_F \quad [3b]$$

$$Z_{LF} = R_p + R_{sol} = \lim_{E_{ac0} \rightarrow 0} \lim_{\omega \rightarrow 0} Z_T \quad [3c]$$

However, the definitions can be generalized in nonlinear regime as^{13–15,22}

$$R_{t,NL}(E_{ac0}) = \lim_{\omega \rightarrow \infty} Z_F \quad [4a]$$

$$Z_{LF,NL}(E_{ac0}) = R_{p,NL}(E_{ac0}) + R_{sol} = \lim_{\omega \rightarrow 0} Z_T \quad [4b]$$

If the actual reaction is a simple electron transfer reaction in the kinetic controlled regime, then the complex plane plot of the impedance spectrum will be a semi-circle and R_t and R_p will be equal. However, in case of an arbitrary electrochemical reaction, R_t and R_p will be different.

Usually, impedance data is analyzed assuming that the investigated system fulfills the conditions of causality, stability and linearity.^{1,3} The conformity of the impedance data to these conditions should be validated using the integral transforms known as Kramers Kronig Transforms (KKT)^{27,32} or their equivalents such as measurement model approach^{33–35} or linear KKT.³⁶ Subsequently, the data is analyzed to obtain physical insights into the investigated system by either fitting suitable electrical equivalent circuits (EEC) or using reaction mechanism analysis (RMA).³ RMA analysis of impedance data aims to identify the mechanism of the physico-electrochemical processes occurring at the electrode electrolyte interface. In this method, several alternate mechanisms are considered and the mechanism and the kinetic parameters which best fit the experimental data are chosen to represent the investigated phenomenon.

Development of NLEIS Analysis Methodology

In the early reports on NLEIS, nonlinear impedance refers to the measurements performed at fundamental frequency when a large amplitude perturbation is applied onto the system.^{11–15,17,19,22} If the impedance is a function of the amplitude of the applied signal, then the data should be analyzed by including the nonlinear terms. However, in some cases one may get significant second or third harmonic but the fundamental impedance may be invariant to the applied amplitude.⁸ Measuring higher harmonics while simultaneously monitoring the dependence of impedance on the applied signal amplitude will enable identification of nonlinearity in the system. Based on the complexity of the system investigated, the studies on NLEIS are organized and summarized below.

Simple electron transfer reaction.—One of the simplest electrochemical reaction is an elementary redox reaction given by



The reaction kinetics are given by the Butler-Volmer equation. If the solution resistance is negligible and if the mass transfer is rapid, the faradaic current can be written as

$$i_F = F(k_{10}e^{b_1E} - k_{-10}e^{b_{-1}E}) \quad [6]$$

where k_{10} and k_{-10} are pre-exponents and the parameters b_1 and b_{-1} are related to the charge transfer coefficient α by $b_1 = \alpha F/(RT)$,

$b_{-1} = -(1-\alpha)F/(RT)$. Here F is the Faraday constant, R is the universal gas constant, T is the temperature and the potential E is measured with respect to the equilibrium potential. If the applied potential deviates significantly from the equilibrium potential, then one of the reactions can be neglected, leading to further simplification of Eq. 6.

Upon application of a sinusoidal potential superimposed on a *dc* potential, the current can be written using modified Bessel function of first kind (I_n) and the nonlinear impedance at any perturbation amplitude and frequency can be written in analytical form.³⁷ It is worth noting that exponential of sinusoidal function is a generating function for $I_n(x)$, as given by³⁸

$$e^{x \sin(\omega t)} = I_0(x) + 2 \sum_{n=0}^{\infty} (-1)^n I_{2n+1}(x) \sin\{(2n+1)\omega t\} + 2 \sum_{n=1}^{\infty} (-1)^n I_{2n}(x) \cos\{2n\omega t\} \quad [7]$$

where

$$I_n(x) = \left(\frac{1}{2}x\right)^n \sum_{m=0}^{\infty} \frac{\left(\frac{x}{2}\right)^{2m}}{m!(n+m)!} = x^n \sum_{m=0}^{\infty} \frac{1}{2^{2m+n} m!(n+m)!} x^{2m} \quad [8]$$

Wong and MacFarlane derived analytical expressions for the first four harmonics of a simple redox reaction with finite layer diffusion and experimentally measured the first and second harmonic response of ferrocene/ferrocenium redox system as well as $K_3[Fe(CN)_6] / K_4[Fe(CN)_6]$ redox systems.³⁹ It should be noted that there is no universally accepted definition of *nonlinear impedance* at higher harmonics.³⁹ On one hand, current at higher harmonics (i_{no}) is related to the n^{th} power of applied perturbation (E_{ac0})ⁿ, at least to the first approximation. However the ratio of (E_{ac0})ⁿ to i_{no} does not have the units of $\Omega\text{-cm}^2$. On the other hand, the ratio E_{ac0}/i_{no} has the correct units of impedance, but is dependent on E_{ac0} even for small values of E_{ac0} (for $n \geq 2$), while the linear impedance is independent of E_{ac0} at small values of E_{ac0} . Hence, most of the NLEIS studies actually report the measured values (current or potential) at higher harmonics, and not impedance. A few articles do define and report impedance at higher harmonics.^{39,40} In this study,³⁹ NLEIS measurements were carried out using a Solartron 1250 FRA and 1186 potentiostat and the theoretical predictions matched well with the experimental results. The second harmonic results also showed that the contribution from double layer capacitance is not significant.

More recently, NLEIS was used to reinvestigate the faradaic process and double layer capacitance of the classical $K_3[Fe(CN)_6] / K_4[Fe(CN)_6]$ redox system.^{8,9} NLEIS measurements were carried out using a potentiostat, a PC based function generator and digitizer (analog to digital converter or ADC). A schematic of experimental set up is shown in the Fig. 3. A three electrode system with gold as working electrode, platinum foil as counter electrode and Ag/AgCl as reference electrode was used in the experiments. The electrolyte used contained varying concentrations of $K_3Fe(CN)_6$ and $K_4Fe(CN)_6$, ranging from 5 mM to 25 mM and the data were acquired in potentiostatic mode. The input amplitude was 100 mV and the frequency range was set from 0.05 Hz – 10 KHz. The odd harmonic currents recorded for a cell with 5 mM of $K_3Fe(CN)_6 / K_4Fe(CN)_6$ at $E_{ac0} = 100$ mV is shown as open symbols in Fig. 4a and the results show that the magnitude of the odd harmonics decrease with increase in harmonic order. The equation representing the faradaic current was expanded in Taylor series up to the 6th term and rearranged into Fourier series. At low frequencies, diffusion will play a significant role and Laplace transform was employed to compute the harmonic currents.⁹ The theoretical predictions are shown as filled symbols in Fig. 4a. Although the match is not quantitative, the simulated harmonic currents show similar trend as the experimentally measured data. The noticeable differences between the simulated and the experimental curves at the high frequency end are likely due to the variation in the solution resistance values in the two cases. In the simulations the solution resistance was deliberately set to zero, whereas in the experiments the double layer capacitance

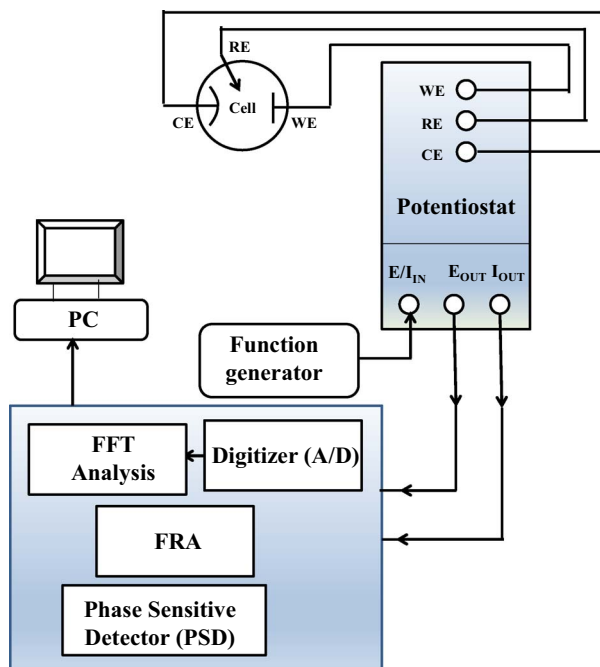


Figure 3. Schematic of experimental setup used to acquire NLEIS. WE – working electrode, RE – reference electrode and CE – counter electrode. Adapted from.^{4,8,9}

and solution resistance contribute significantly to the total current in the high frequency region resulting in the decrease in the current magnitude. The odd harmonics together with the even harmonics (Fig. 4b) were analyzed to calculate the charge transfer coefficient. While this analysis methodology can be applied to simple electron transfer reaction, extending it to complex reactions will be challenging.

If the electrode-electrolyte interface is represented by a simple capacitor (Helmholtz model), then the corresponding current response will not have any higher harmonic component. If the system is actually better represented by Stern model, which is a combination of Helmholtz and Gouy-Chapman model, and if NLEIS is acquired at point of zero charge (PZC), then odd harmonics would be nonzero, but even harmonics would be zero due to the symmetry around PZC.⁸ Interestingly, at high frequencies, impedance at fundamental frequency did not show any dependence on E_{ac0} even up to 100 mV, but significant higher harmonics were present. This shows that invariance of linear EIS cannot be always used as a validation of system linearity.⁸ When the dc potential is moved away from PZC, both even and odd harmonics would have contributions from double layer capacitance element.⁸ Experimental results show that when the measurement is done near OCP, at low frequencies, even harmonics are lower in magnitude compared to the next odd harmonics, i.e. $|i_{2\omega}| < |i_{3\omega}|$ and $|i_{4\omega}| < |i_{5\omega}|$ (results not shown here). When the symmetry is broken by applying a dc bias (± 100 mV), the magnitude of even harmonics $|i_{2n\omega}|$ was greater than the corresponding values of next odd harmonics $|i_{(2n+1)\omega}|$, as seen in Fig. 4b.⁹ Unlike traditional EIS where multiple measurements are used to calculate the charge transfer coefficient, it can be estimated by analyzing NLEIS in a single measurement for this reaction.⁹

Hernandez-Jaimes et al. derived nonlinear transfer functions to model the nonlinear EIS responses of gelatinized starch suspensions (GSS), commonly used as biopolymeric electrolytic system.⁴¹ The main focus of the study was to develop a mathematical analysis for NLEIS and hence a simple system was chosen. The impedance response at fundamental frequency to large amplitude perturbation was recorded using PARSTAT 2263. The EEC corresponding to the nonlinear form of Cole-Cole equation is shown in Fig. 5. The nonlinear constant phase element (NL-CPE) pre-exponent and the high and low

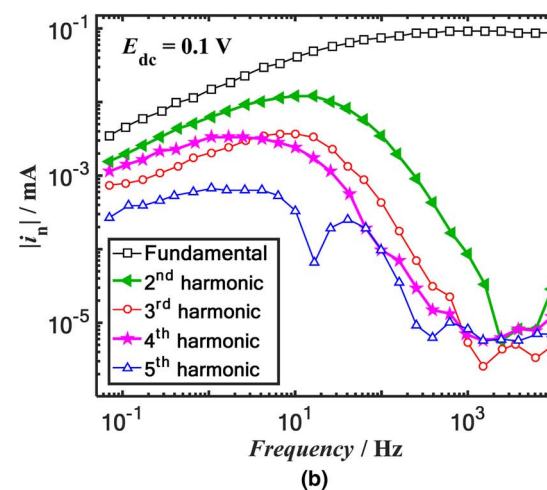
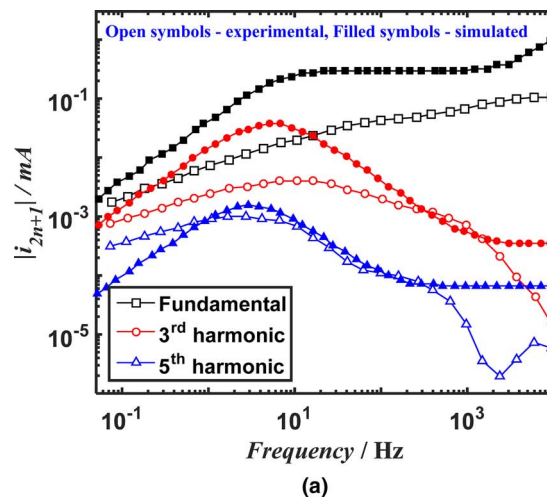


Figure 4. (a) Odd harmonic currents of Au electrode in 5 mM Ferricyanide and 5 mM Ferrocyanide. Open symbols represent the experimental data while filled symbols represent simulated data with zero solution resistance (b) Experimental NLEIS using Au pseudo reference electrode on a $K_4[Fe(CN)_6]$ 5 mM / $K_3[Fe(CN)_6]$ 5 mM solution cell under dc bias. Adapted from Ref. 9.

frequency limit of the impedance values were used to estimate an equivalent capacitance (C). EEC element values were correlated to the perturbation amplitude and it was shown that R_p and C could be related exponentially to E_{ac0} , with R_p decreasing with E_{ac0} and C increasing with E_{ac0} . The CPE exponent also showed a dependence on E_{ac0} but it was not discussed in detail. It is worth noting that the CPE parameters are intrinsically coupled and are not expected to vary independently.⁴² The impedance of the nonlinear EEC consisting of two nonlinear resistors (NL- R_{sol} and NL- R_p) and a nonlinear CPE (NL-CPE) was obtained by expressing current as a nonlinear function of the applied potential. Using Kirchhoff's law, nonlinear EEC shown

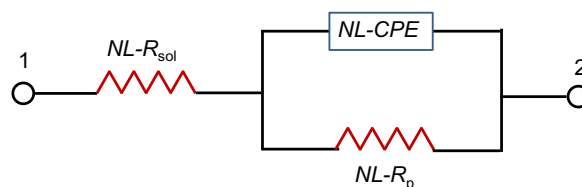


Figure 5. Nonlinear circuit scheme used to represent gelatinized starch suspension. NL- R_{sol} and NL- R_p – E_{ac0} dependent nonlinear resistances. NL-CPE – nonlinear constant phase element. Adapted from Ref. 41.

in Fig. 5 was represented by a system of nonlinear differential equations. Then the expressions for impedance response functions were written by employing Fourier series expansion and equating the first harmonic terms. This procedure was applied to model the impedance of GSS and the nonlinear model describing exponential decrease of the low frequency resistance with increase in amplitude was found to give best match with the experimental results. Higher harmonics were not recorded or included in the model development. While the exponential dependence of the circuit element values could model the data well, it is difficult to extract physical insight into the reaction process and it is not clear if the model could predict the higher harmonics correctly.

When mass transfer is rapid but R_{sol} is not negligible, the governing equations can be solved numerically to calculate the impedance spectra at any frequency.^{43–46} Kramers Kronig Transforms (KKT) are mathematical equations that can be used to confirm that the response arise from a system which is linear, causal and stable. While KKT are successful in flagging violations of stability,⁴⁷ they are only partially successful in identifying nonlinearity effects.^{27,44} If R_{sol} was negligible, then for a simple electron transfer reaction, KKT will not flag nonlinearity effects. However, if R_{sol} was considerably high and if the data is acquired in a wide frequency range containing transition frequency given by

$$f_t = \frac{1}{2\pi C_{dl}} \left(\frac{1}{R_{sol}} + \frac{1}{R_{t,NL}} \right) \quad [9]$$

where C_{dl} is the double layer capacitance, the nonlinearity will be detected by KKT.^{43,44} Since the analysis was based on simple electron transfer reaction, these conclusions may not be applicable to more complex reactions.

Estimation of polarization resistance.—The current potential relationship of a diode is very similar to that of a simple irreversible electron transfer reaction and hence, in early studies, a circuit with a diode as nonlinear element has been employed to understand NLEIS response of such systems.^{12–15,17} Based on NLEIS measurements on electrical circuits with a diode, and by expanding the current potential relationship in Taylor series up to third order terms, Darowicki proposed a parabolic relationship between $(R_{p,NL})^{-1}$ and E_{ac0} .^{12,17} This can be used to estimate R_p , i.e. the polarization resistance at vanishingly small perturbation amplitudes. The polarization resistance measured using NLEIS matched with the value calculated using steady state current potential technique. This is applicable when the solution resistance is negligible, diffusion limitations are not present and the perturbation amplitude is moderate.²¹

If the impedance is acquired under pseudo potentiostatic conditions, when R_{sol} is significant and perturbation amplitude is large, the potential drop across electrode-electrolyte interface will not be sinusoidal even if the potential applied across the entire system is sinusoidal.^{13–15,46} Initially, it was thought that $R_{p,NL}$ in these cases will always decrease with E_{ac0} , but it was later shown that $R_{p,NL}$ may increase or decrease with E_{ac0} depending on the value of bias potential (E_{dc}) and R_{sol} .^{13–15} Expressions were also derived for $R_{p,NL}$ for electrochemical reactions obeying Tafel, or Butler Volmer or Nernst law or an electron transfer reaction coupled with electrochemical adsorption reaction (E-EAR), when NLEIS was measured under potentiostatic or galvanostatic conditions.^{13–15} In corrosion studies, it is desirable to measure the linear polarization resistance (R_p) quickly and independently of polarization experiments.¹⁶ NLEIS method was employed to determine the charge transfer coefficient of the oxygen evolution reaction on a composite material (ethylene-propylene terpolymer combined with type P-1250 acetylene carbon black) in 0.05 M Na_2SO_4 , which was used as anode in cathodic protection.¹⁶ Impedance spectra were recorded at three different values of effective potentials of $E_{dc} = 1.186$ V, $E_{dc} = 1.277$ V and $E_{dc} = 1.357$ V, using $E_{ac0} = 30, 150, 180, 210, 240, 280$, and 340 mV. Analysis of NLEIS data enabled determination of the kinetic coefficient values of the investigated system in a shorter time when compared to potentiodynamic polarization technique.

Theoretical analysis was performed to predict the nonlinear response of a corrosion process described by Butler Volmer kinetics¹⁷ and the amplitude dependency of the polarization resistance was derived as:

$$(R_{p,NL})^{-1} = i_{corr} \left[(b_a + b_c) + \frac{(b_a^3 + b_c^3)}{8} E_{ac0}^2 + \frac{(b_a^5 + b_c^5)}{192} E_{ac0}^4 + \dots \right] \quad [10]$$

where i_{corr} is the corrosion current, $b_a = \frac{2.3}{b_{aT}}$ and $b_c = \frac{2.3}{b_{cT}}$ and b_{aT} and b_{cT} are the anodic and cathodic Tafel coefficients respectively. By measuring the NLEIS at various amplitudes and calculating the $R_{p,NL}$ as a function of E_{ac0} , one may calculate i_{corr} , but since the expression is symmetrical with respect b_a and b_c , they cannot be obtained unambiguously without additional experiments.⁴⁸ For a general current potential relationship given by $i = \text{function}(E)$, an accurate relationship was derived by Diard et al. to relate $R_{p,NL}$ to the applied perturbation, in presence of significant solution resistance²²

$$(R_{LF,NL}) = (R_{p,NL} + R_{sol}) = \left(\sum_{m=0}^{\infty} [2^{2m} m! (m+1)!]^{-1} \times \left(\frac{d^{2m+1} i}{dE^{2m+1}} \right) E_{ac0}^{2m} \right)^{-1} \quad [11]$$

This equation was employed to estimate the error in R_p , when a large amplitude perturbation is employed.¹³ Experiments performed with the classical $\text{K}_3[\text{Fe}(\text{CN})_6]/\text{K}_4[\text{Fe}(\text{CN})_6]$ redox couple on Pt electrode showed that $R_{p,NL}$ may decrease or increase with E_{ac0} , depending on the applied dc bias (E_{dc}) during NLEIS acquisition. This contrasts with the earlier report which predicted that $R_{p,NL}$ will always decrease with E_{ac0} .¹⁷ For an electrochemical system obeying Butler Volmer equation, the current at fundamental and higher harmonics can be written in terms of modified Bessel functions, I_n .⁴⁸ This analysis was extended to a reaction where the anodic reaction was kinetics limited and the cathodic reaction was under kinetic and diffusion control.⁴⁹ At low frequency perturbation, if the first three harmonics, along with the phase at fundamental frequency are measured, then the corrosion current and Tafel parameters can be estimated, provided the diffusivity and surface concentration of the oxidant are known.⁴⁹

Employing operator notation for the derivative as $D_E = d/dE$, the following identity can be established from Eq. 8.

$$[I_1(E_{ac0} D_E)](i) = \left[\left(\frac{1}{2} \right) \sum_{m=0}^{\infty} \frac{(E_{ac0} D_E)^{2m+1}}{2^{2m} m! (m+1)!} \right] (i) = \frac{E_{ac0}}{2} \sum_{m=0}^{\infty} \frac{(E_{ac0})^{2m}}{2^{2m} m! (m+1)!} \frac{d^{2m+1} i}{dE^{2m+1}} \quad [12a]$$

Then, Eq. 11 can be re-written in a concise form as

$$(R_{LF,NL})^{-1} = (R_{p,NL} + R_{sol})^{-1} = \frac{2}{E_{ac0}} [I_1(E_{ac0} D_E)](i) \quad [12b]$$

This follows from the fact that the Taylor series expansion

$$i_{dc+ac} = f(E_{dc} + E_{ac0} \sin(\omega t)) = \sum_{m=0}^{\infty} \frac{f^{(m)}(E)|_{E_{dc}} [E_{ac0} \sin(\omega t)]^m}{m!} = \sum_{m=0}^{\infty} \frac{D_E^m(i) [E_{ac0} \sin(\omega t)]^m}{m!} \quad [12c]$$

can be written as

$$i_{dc+ac} = [e^{E_{ac0} \sin(\omega t) D_E}] (f(E)) = [e^{E_{ac0} \sin(\omega t) D_E}] (i) \quad [12d]$$

At low frequency limit, in general, for odd 'n' the n^{th} harmonic current term can be written as

$$i|_{n\omega} = 2(-1)^{\left(\frac{n-1}{2}\right)} \sin(n\omega t) \times \{[I_n(E_{ac0}D_E)](i)\} \quad [13]$$

and for even 'n' ($n > 0$), the n^{th} harmonic current term can be written as

$$i|_{n\omega} = 2(-1)^{\frac{n}{2}} \cos(n\omega t) \times \{[I_n(E_{ac0}D_E)](i)\} \quad [14]$$

The first term corresponding to $m = 0$ in Eq. 11 would be equal to $(R_p + R_{sol})^{-1}$. However, the actual mechanism and theoretical current-potential relationship may not be known and in fact, often the aim of the experiments and analyses is to unravel the kinetics. In theory, it is possible to compute numerical derivatives of steady state current potential curves, but the accuracy would be very poor, especially for higher order derivatives. Hence practically it can be very challenging to estimate R_p using the above equations. As an alternative, if the higher harmonics of current density are measured, then even without knowing the current potential relationship of the system, it is possible to estimate R_p using the odd harmonics from a single input perturbation,¹⁵ as given in Eq. 15.

$$R_p + R_{sol} = \frac{E_{ac0}}{\sum_{n=0}^{\infty} (2n+1) i_{2n+1}(\omega \rightarrow 0)} \quad [15]$$

It is worth noting that higher harmonic currents tend to have low magnitude and hence the accuracy in the estimate is good even when the terms are limited to 5th harmonic. This was illustrated by NLEIS measurement on a circuit with a diode, as well as on the classical $K_4[Fe(CN)_6]/K_3[Fe(CN)_6]$ redox couple on Pt electrode.¹⁵ Kiel et al. employed a similar approach on reactions following Butler Volmer kinetics, to propose a criterion to verify linearity in EIS measurements and to extract the effective linear EIS data from nonlinear EIS measurements.⁵⁰ Another approach to estimate $R_{p,NL}$ for Tafel kinetics with significant R_{sol} employs Lambert-W functions to get closed form solutions.⁵¹

Under pseudo galvanostatic conditions, upon application of a current perturbation of amplitude i_{ac0} , and decomposition of the resulting potential into Fourier series, the polarization resistance can be written as¹⁵

$$R_{p,NL} + R_{sol} = (R_p + R_{sol}) \left[1 + \sum_{n=1}^{\infty} \frac{d^{2n+1}E/di^{2n+1}}{2^{2n}n!(n+1)! \left(\frac{dE}{di}\right)} (i_{ac0})^{2n} \right] \quad [16]$$

which can be written in a concise form as

$$\begin{aligned} R_{p,NL} + R_{sol} &= \frac{2(R_p + R_{sol})}{\left(\frac{dE}{di}\right) \times i_{ac0}} [I_1(i_{ac0}D_i)E] \\ &= \frac{2(R_p + R_{sol})}{R_{p,NL} \times i_{ac0}} [I_1(i_{ac0}D_i)E] \end{aligned} \quad [17]$$

where $D_i = d/di$. In case the current potential relationship is not known, but higher harmonics of the potential to large amplitude current perturbations are measured in pseudo galvanostatic mode, the polarization resistance can be estimated as

$$R_p + R_{sol} = \frac{\sum_{n=0}^{\infty} (2n+1) E_{2n+1}(\omega \rightarrow 0)}{i_{ac0}} \quad [18]$$

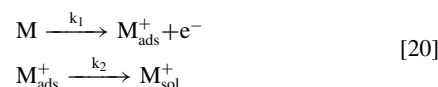
The Equations 11–18 are applicable to any general non-blocking electrode.

Complex reactions.—Nonlinear current response of a two-step electrode reaction with an adsorbed intermediate, shown in Eq. 19, was derived using Taylor series expansion method.⁵²



Here A is the reduced species, C is the oxidized species and $(B)_{ad}$ is the adsorbed intermediate species. The expressions for high frequency charge transfer resistance, polarization resistance (described as low frequency charge transfer resistance) and relaxation time of the intermediate species, under nonlinear conditions were obtained. For systems with significant electrolytic resistance, the effective perturbation amplitude was found to vary with frequency, with the maximum amplitude at low frequency and the minimum amplitude at high frequency. In addition, an expression to compute the maximum perturbation amplitude which can ensure the linear behavior of the investigated system was presented.¹⁹ Diard et al. developed expressions for the nonlinear impedance at the fundamental frequency for a similar two-step reaction with adsorbed intermediate.²³ It was shown that the nonlinear behavior of the investigated system depends on kinetic parameters, electrode potential, input perturbation amplitude and frequency and suggested corrections to the equations proposed in Ref. 52. The nonlinear charge transfer resistance $R_{t,NL}$ was also calculated using a fairly complex expression and compared with the values obtained using numerical methods.²³ The governing equations can be expanded in Fourier series instead and the current can be expressed as Fourier series in terms of modified Bessel function, I_n .³⁷ In the case of reactions with adsorbed intermediate, the surface coverage of the adsorbed intermediate was assumed to vary periodically and contain higher harmonics. The differential equation for surface coverage of the adsorbed intermediate (θ) was reduced to a system of algebraic equations to solve for the ac components of θ , which are then used to solve for the components of the current analytically. Thus, the modified Bessel functions enabled easy and more accurate determination of harmonics than the Taylor series expansion method.

In general, it is challenging to derive analytical expressions for the nonlinear impedance, and therefore numerical methods were sought. By employing direct numerical simulations, the nonlinear impedance spectra a similar reaction with an adsorbed intermediate species, described in Eq. 20, was calculated for Langmuir isotherm or Frumkin adsorption isotherm model.⁴⁵



When $R_{sol} \approx 0$ and mass transfer is rapid, for Langmuir isotherm model, the steady state surface coverage of adsorbed intermediates is

$$\theta_{M_{ads}^+}^{ss} = \frac{(1/k_{2DC})}{\sum_{i=1}^2 (1/k_{iDC})} \quad [21]$$

Here, k_{iDC} represents the rate constant at the dc potential (E_{dc}). Under large amplitude sinusoidal perturbation of high frequency ($\omega \gg 1$) and at long times, the surface coverage will oscillate around

$$\theta_{M_{ads}^+}^{AV} = \frac{1/(k_{2DC}I_0(b_2E_{ac0}))}{\sum_{i=1}^2 1/(k_{iDC}I_0(b_iE_{ac0}))} \quad [22]$$

i.e. the usual expression to find the steady state average can be employed with a correction factor in terms of I_0 . The study was extended to unstable systems as well and the results showed that KKT detects violations of stability but not linearity. Numerical simulations also show that when a sinusoidal perturbation is applied, the system reaches steady state only after some time (Fig. 6a), and acquiring spectrum before it reaches steady state will result in incorrect impedance spectrum⁴⁷ as shown in Fig. 6b. The faradiac impedance in the mid frequency region was distorted due to the surface coverage drift of the adsorbed intermediate, especially when the perturbation amplitude was large. Hence EIS data must be recorded after allowing sufficient time for the surface coverage values to settle. The drift effect is more pronounced when the R_{sol} is high⁴⁶ and this was illustrated for reaction with two adsorbed intermediates, exhibiting a variety of patterns in

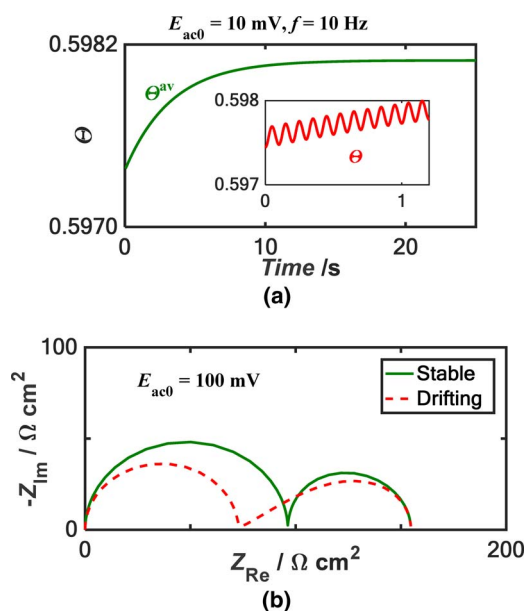


Figure 6. For the reaction mechanism described in Eq. 20, (a) transient and average surface coverage of the adsorbed intermediate (M_{ads}^+) as a function of time for potential perturbation at 10 Hz. The inset shows the transient surface coverage, at the initial stage. (b) Complex plane plots of impedance spectra at stable and drifting conditions. Adapted from Ref. 47.

complex plane plots. Using a heuristic approach, an estimate for $R_{t,NL}$ for the reaction in Eq. 20 was given as²⁷

$$R_{t,NL} = \left[F \left\{ \frac{2I_1(b_1 E_{ac0})}{E_{ac0}} k_{1DC} (1 - \theta_{ads}^{AV}) + \frac{2I_1(b_2 E_{ac0})}{E_{ac0}} k_{2DC} \theta_{ads}^{AV} \right\} \right]^{-1} \quad [23]$$

which is essentially the same expression used for R_t , but with a correction factor in terms of I_1 for the rate constants, and the average surface coverage values given by Eq. 22 should be used instead of the steady state surface coverage values.

Using the proposed numerical simulation method, the nonlinear impedance of other complicated reactions can also be predicted.^{27,46,47} In addition to calculating the values at fundamental frequency, the higher harmonics can easily be computed. NLEIS data were simulated numerically to understand the capability of KKT to detect nonlinearities in the data.²⁷ A reaction with an adsorbed intermediate species, an electrochemical - chemical reaction with two adsorbed intermediates and a reaction with two adsorbed species and showing negative resistance were examined. The results suggest that KKT could detect the nonlinearities successfully, if logarithm of faradaic current vs. dc potential exhibited nonlinear relationship.

Harmonic analysis of ionic transport phenomena through ion exchange membrane systems was performed using numerical methods.⁵³ A theoretical model based on Nernst-Planck equation for dilute solutions was employed to compute the nonlinear response of ion transport through a cation exchange membrane system. The system was perturbed with a sinusoidal current signal and the resulting potential was expanded in Fourier series and resolved into harmonic components. The result at fundamental and higher harmonic, at different dc values, were analyzed and the low frequency impedance was observed to increase with the input amplitude.

Applications of NLEIS

Most of the above studies are restricted to predicting the nonlinear response for a known electrochemical kinetics, and in most cases it was also assumed that the mass transfer is rapid. Usually the kinetics are not known and EIS is often employed to understand the physicochemical processes of the electrochemical system. In ideal cases, ex-

perimental data such as potentiodynamic polarization and traditional EIS as well as NLEIS would be acquired, and a detailed mechanism with a particular set of kinetic and transport parameters would be proposed and the predicted values would be compared with the experimental data. A good match between the simulated and experimental results is taken as validation of the proposed mechanism. Many a times, this is a very challenging task and it is difficult to achieve even a semi-quantitative match. If a quantitative match cannot be achieved, then even identifying a correlation between the physical phenomena and trends in the measured data can offer some insights into the system. Literature reports of NLEIS results which help in identifying the physical processes, and published in the last decade, are summarized below.

Solid oxide fuel cells.—Electrochemical impedance of oxygen reduction reaction (ORR) at porous mixed conducting electrode was modeled using a physical model based on bulk transport pathway.^{54,55} This model was extended to incorporate the nonlinear effects due to the thermodynamic relation between the mole fraction of oxygen vacancies and oxygen activity in the solid and due to the oxygen exchange kinetics at the gas solid interface. The governing reaction mechanism in SOFC cathode material Lanthanum Strontium Cobalt Oxide ($\text{La}_{1-x}\text{Sr}_x\text{CoO}_{3-\delta}$) was investigated using NLEIS in galvanostatic mode.^{4,5,56,57} Wilson et al. studied the oxygen reduction mechanism at SOFC cathode using symmetric cathode cells.⁴ They demonstrated the experimental measurement and analysis of first and third harmonic components of the voltage response of porous $\text{La}_{0.8}\text{Sr}_{0.2}\text{CoO}_{3-\delta}$ (LSC-82) on samaria doped ceria (SDC) at 750 °C in air⁴ under galvanostatic mode. NLEIS data were recorded using potentiostat/galvanostat along with a function generator and digitizers, using an experimental setup similar to the one shown in Fig. 3. Gaussian apodization, also known as windowing, was employed before subjecting the time domain data to Fourier analysis. The n^{th} harmonic potential was written as (in our notation)

$$E_n = \sum_{r=1}^{\infty} \left(\frac{i_{ac0}}{i_{ac0,NL}} \right)^{n+2r-2} E_{n,n+2r-2}(\omega) \\ = (i_{ac0})^n \sum_{r=0}^{\infty} \frac{E_{n,n+2r}(\omega)}{(i_{ac0,NL})^{2r+n}} (i_{ac0})^{2r} \quad [24]$$

Here, $E_{n,n+2r}$ is amplitude independent *primary response coefficient* and $i_{ac0,NL}$ is the current density that marks the onset of nonlinear effects in the system. Note that the equation is structurally similar to the series expansion of the modified Bessel function, given in Eq. 8. For practical applications, Eq. 24 can be truncated after a few terms. The authors experimentally measured the potential up to 3rd harmonics at varying amplitudes and calculated the primary response coefficients ($E_{n,n+2r}$) by fitting the harmonics to the power series in Eq. 24, for each frequency. The major driving force of oxygen exchange kinetics was assumed to be oxygen chemical potential difference between the gas and solid bulk. Model parameters corresponding to bulk thermodynamic properties of LSC-82 and $\text{La}_{0.43}\text{Sr}_{0.57}\text{CoO}_{3-\delta}$, along with possible oxygen adsorption kinetics were evaluated to obtain the best match with the experimentally measured first and third harmonic data. Based on the analysis of nonlinear EIS data, in particular the third order spectra ($E_{1,3}$ and $E_{3,3}$), the authors concluded that the previously proposed model assuming complete bulk transport and thermodynamic properties of LSC-82 could not explain the higher harmonic results satisfactorily. Hence the authors suggested that the model has to be modified to include combined surface and bulk pathways, where the concentration of the vacancies is higher at the surface than in the bulk.

In a related work, the mechanism of oxygen reduction on thin film $\text{La}_{0.6}\text{Sr}_{0.4}\text{CoO}_{3-\delta}$ and the rate limiting mechanism in porous $\text{La}_{0.6}\text{Sr}_{0.4}\text{CoO}_{3-\delta}$ (LSC 6410) electrodes on a gadolinium doped ceria $\text{Ce}_{0.9}\text{Gd}_{0.1}\text{O}_{1.95}$ (GDC) electrolyte were investigated using NLEIS.^{5,57} Various mechanisms of oxygen reduction on thin films were evaluated using the NLEIS data.⁵ In one case, it was assumed that (i) energy

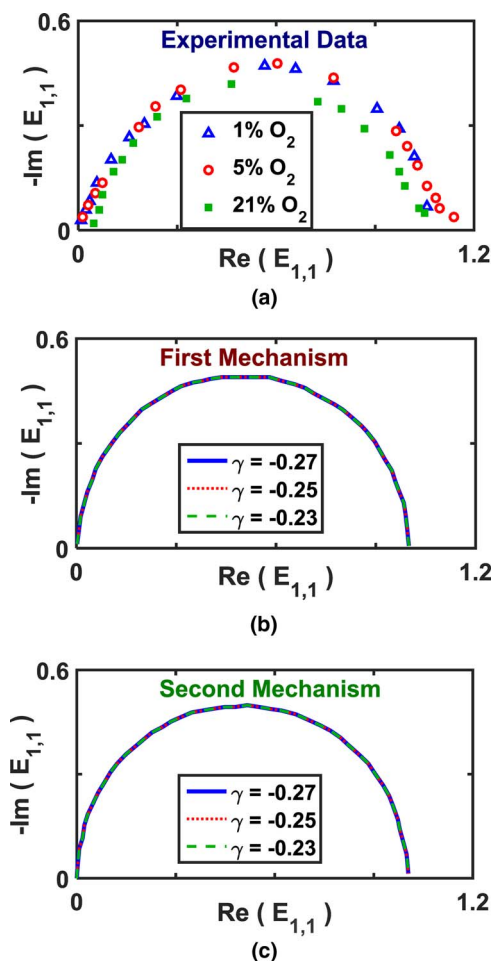


Figure 7. Complex plane plot of response of primary response coefficient at fundamental frequency ($E_{1,1}$) (a) experimental data under different O_2 gas environments in diluent N_2 at 725°C . Calculated data as a function of thermodynamic factor γ for (b) first mechanism (c) second mechanism. Both mechanisms predict $E_{1,1}$ equally well. The lines at various γ overlap in (b) and (c) and are practically indistinguishable. Adapted from Ref. 5.

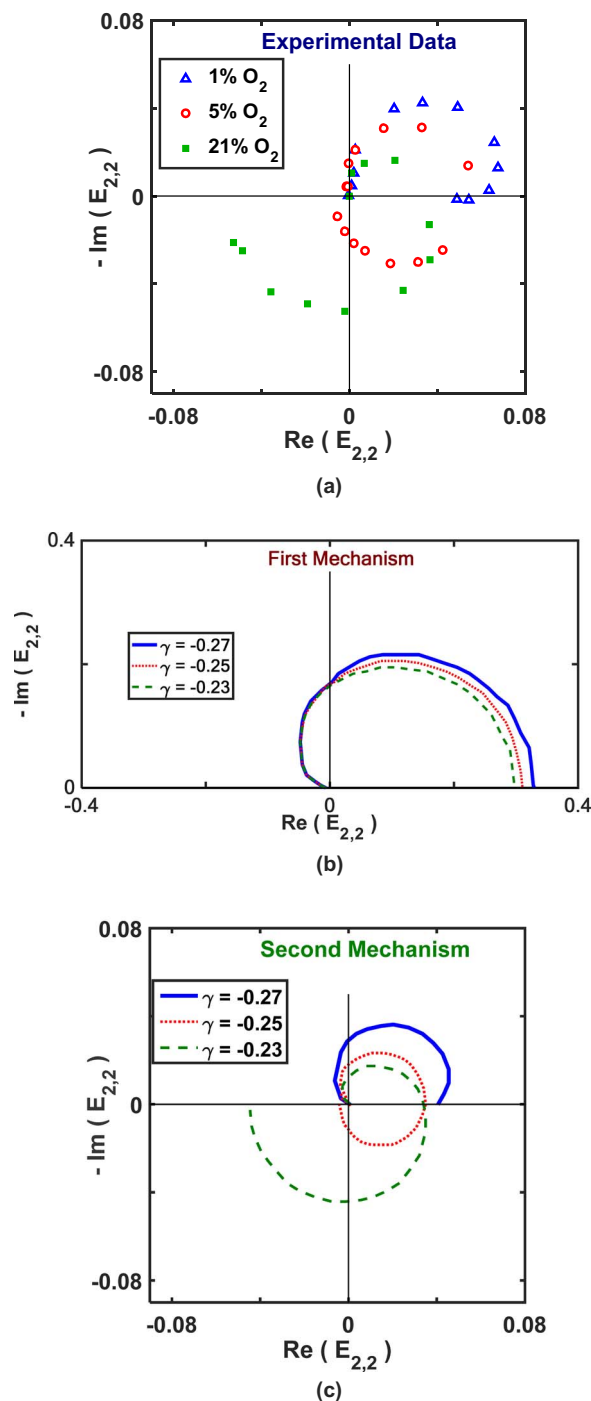


Figure 8. Complex plane plot of the primary response coefficient at 2nd harmonic ($E_{2,2}$) (a) experimental data under different O_2 gas environments in diluent N_2 at 725°C . Calculated data as a function of thermodynamic factor γ for (b) first mechanism and (c) second mechanism. Note that the predictions of first mechanism do not match the experimental data, while the predictions of second mechanism match them well. Adapted from Ref. 5.

barrier to dissociation was the rate limiting step. In the second case when the rate limiting step was assumed to involve (ii) dissociative adsorption of O_2 on the vacant surface oxygen sites without an energy barrier, a match could be found. The impedance values in normalized form are presented in Fig. 7 as complex plane plots. Fig. 7a shows the experimental results while Figs. 7b and 7c show the predictions of mechanisms 1 and 2 respectively. Clearly, both mechanisms yielded similar impedance values and thus linear EIS cannot be used for model discrimination in this case. The experimentally measured and model predicted 2nd harmonics are presented as complex plane plots in Fig. 8. A comparison of Figs. 8a and 8b show that the first model predictions of second harmonic deviate significantly from the experimental data. On the other hand, the match between experimental results (Fig. 8a) and second model results (Fig. 8c) are good. A comparison of third harmonic results (not shown here) also led to the same conclusion. Thus NLEIS data help identify the correct mechanism in this study. In addition to the above two models, a few other mechanisms were also tested, viz. (iii) molecular non-dissociative adsorption or (iv) dissociation of other charged intermediates or (v) change in oxidation state of adsorbed monatomic intermediates or (vi) exchange of oxygen vacancies between surface and bulk were considered as rate limiting and evaluated. However, in all the above cases, the predicted trends were inconsistent with the experimental NLEIS results and hence these mechanisms were rejected.

NLEIS data measurement and analysis for porous $\text{La}_{0.6}\text{Sr}_{0.4}\text{Co}_{3-8}$ (LSC 6410) electrodes on a gadolinium doped ceria $\text{Ce}_{0.9}\text{Gd}_{0.1}\text{O}_{1.95}$ (GDC) electrolyte indicate that a combined bulk vacancy and surface interstitial diffusion pathway and a modification of the material surface due to Sr segregation have to be included in the existing model in order to obtain good fit for the 2nd and 3rd harmonics.⁵⁷ The drift effects in LSC electrode characteristics were also investigated using similar NLEIS approach.⁵⁶ The low frequency impedance values, as a func-

tion of time, were measured on 45 nm thick LSC-82 film at 600 °C and 0.01 atm O₂. Over a period of 18 hours, the low frequency impedance values decreased with time, indicating nonstationary effects in the investigated system. But the second and the third harmonics were not affected much by the drift. Hence the authors concluded that the transient behavior was primarily due to the changes in the rate coefficients rather than any alteration in the mechanism or thermodynamic properties. Furthermore, the effects of Sr segregation in thin film LSC electrodes were studied using the second and third harmonics.⁵⁶ Thus, using NLEIS, a dissociative adsorption based reaction mechanism was deduced for oxygen reduction in varying composition of LSC electrodes in SOFC.

Polymer electrolyte membrane fuel cells.—Apart from the Fourier expansion presented in Eq. 1 and 2, other types of series expansions have also been employed to analyze nonlinear systems. Nonlinear Frequency Response Analysis (NFRA) analysis based on Volterra series and generalized Fourier transform is another approach that can be employed for the investigation of weakly nonlinear systems.^{6,7,58} Volterra expansion is a functional Taylor series,⁵⁹ and the nonlinear response of an electrochemical system to input signal can be expressed in terms of Volterra series. Experimentally, when a time domain signal is processed with a frequency response analyzer (FRA), the frequency domain information (i.e. Fourier series coefficients and phase values) would be extracted. The Fourier series terms, referred to as quasi response functions in the original publication,⁶ can be related to the higher order frequency response functions (HFRF) computed from Volterra series expansion. Under pseudo galvanostatic conditions with an applied perturbation amplitude of i_{ac0} , the relationship can be written (in our notation) as

$$E_1 = i_{ac0} \left[H_1 + \frac{3}{4}(i_{ac0})^2 H_3 + \frac{5}{8}(i_{ac0})^4 H_5 + \dots \right] \quad [25a]$$

$$E_2 = (i_{ac0})^2 \left[\frac{1}{2} H_2 + \frac{1}{4}(i_{ac0})^2 H_4 + \dots \right] \quad [25b]$$

$$E_3 = (i_{ac0})^3 \left[\frac{1}{4} H_3 + \frac{5}{16}(i_{ac0})^2 H_5 + \dots \right] \quad [25c]$$

where H_n is the n^{th} order FRF. Here again, the structural similarity between Eq. 25 and Eq. 8 and 24 is striking. In Eq., 25 if i_{ac0} is selected such that the 3rd and higher harmonics are negligible, then it can be simplified as

$$H_1 = \frac{E_1}{i_{ac0}} \quad [26a]$$

$$H_2 = \frac{2E_2}{(i_{ac0})^2} \quad [26b]$$

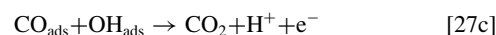
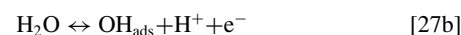
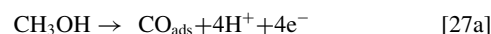
Using these relations, the nonlinear frequency response functions can be calculated from the experimental higher harmonic values. The main advantage of employing HFRF is that they are amplitude independent, while the Fourier coefficients are amplitude dependent. The higher order functions can be derived analytically from a nonlinear model using Volterra series and generalized Fourier transform. A limitation of this particular implementation is that the i_{ac0} cannot be arbitrarily increased to enhance the signal to noise ratio, since the 3rd and higher harmonics would also increase and cannot be neglected. It is possible to use the Eq. 25 instead of the simpler version given in Eq., 26 and the measurements have to be performed at multiple perturbation amplitudes to extract HFRF.

Nonlinear EIS data acquisition and analysis was employed to distinguish between membrane dehydration, flooding and anodic CO poisoning in polymer electrolyte membrane (PEM) fuel cells.⁶ The measurements were performed using potentiostat/galvanostat with Newton 4th phase sensitive analyzer (PSA) PSM 1735. The data were recorded in the galvanostatic mode at three different dc bias currents,

viz. 3A, 6A and 9A corresponding to the activation controlled regime, membrane resistance controlled regime and mass transfer controlled regime respectively. The data were acquired only at single amplitude selected such that the third and the higher harmonics are negligible compared to the first and the second harmonics. The first and second harmonic responses of PEM fuel cell under normal, dehydration and flooding conditions were recorded at $i_{ac0} = 58 \text{ mA cm}^{-2}$, while the experiments to compare the results with and without CO poisoning were conducted at $i_{ac0} = 96 \text{ mA cm}^{-2}$. The NFRA spectra were computed from the experimentally measured harmonic values using Eq. 26. The data obtained at $i_{dc} = 0.231 \text{ A cm}^{-2}$ is shown in Fig. 9. Using normal EIS data, dehydration and CO poisoning cannot be distinguished but the signature of flooding in the EIS was clearly different. On the other hand, the effect of CO poisoning on the 2nd order FRF was distinguishable from other effects.^{6,24} Taken together, 1st and 2nd order FRF can be used to identify flooding, dehydration and CO poisoning unambiguously. Thus, even without any model development, a correlation between physical phenomena and results at 1st and 2nd harmonics could be established using NLEIS data.

The anodic CO poisoning mechanism was investigated using steady state polarization curve, EIS and NLEIS methods.⁷ The experiments were carried out in galvanostatic mode using potentiostat/galvanostat and a FRA. The cell was operated as differential H₂/H₂ cell in order to avoid the masking effects due to the cathodic oxygen reduction reactions on the CO poisoning mechanism. The data was obtained at a dc current value of 0.1 A cm^{-2} using an amplitude of 90 mA cm^{-2} in the experiments without CO poisoning and an amplitude of 60 mA cm^{-2} in the experiments with 20 ppm and 100 ppm CO in the feed gas. These amplitude values were chosen such as the response contains contribution from the fundamental and second harmonics only. The range of frequency employed was from 0.1 Hz to 10 KHz. Bode plot representation of the first and second harmonic responses without CO, 20 ppm CO and 100 ppm CO content in the anodic feed gas are shown in Fig. 10. The mechanism proposed by Springer et al.⁶⁰ was employed and results of FRF H_1 and H_2 were compared with simulated values. The 1st order response (H_1) showed an increase in the magnitude of the low frequency impedance due to the blockage of the active catalyst sites by CO (Fig. 10). In the absence of CO, the magnitude of the 2nd order response (H_2) was found to be negligible and remain more or less constant. With an increase in the CO content, the magnitude of the H_2 shows a significant increase in the low and mid frequency regions. The phase angle of the second order responses also exhibits a considerable shift as the CO concentration in the feed gas increase. Although the match was not quantitative, the trends were captured well.

Direct methanol fuel cells.—Bensmann et al. performed theoretical analysis of methanol oxidation kinetics in direct methanol fuel cell (DMFC).⁵⁸ The reaction studied was



The adsorption model could be Langmuir or Frumkin and the rate constant in Eq. 27c could be potential dependent or independent. All four possible combinations were evaluated by analytically deriving the FRF up to the 2nd order, i.e. H_1 and H_2 .⁵⁸ The main characteristic feature of the first order response function, which is the same as normal EIS, was the appearance of pseudo inductive loop at the low frequency end. All the four kinetic models predict roughly similar fundamental harmonic response and hence model discrimination based on EIS data alone is not possible. The 2nd order responses predicted by the four mechanisms exhibited qualitative differences, especially in the mid frequency region. Therefore it was concluded that measurement

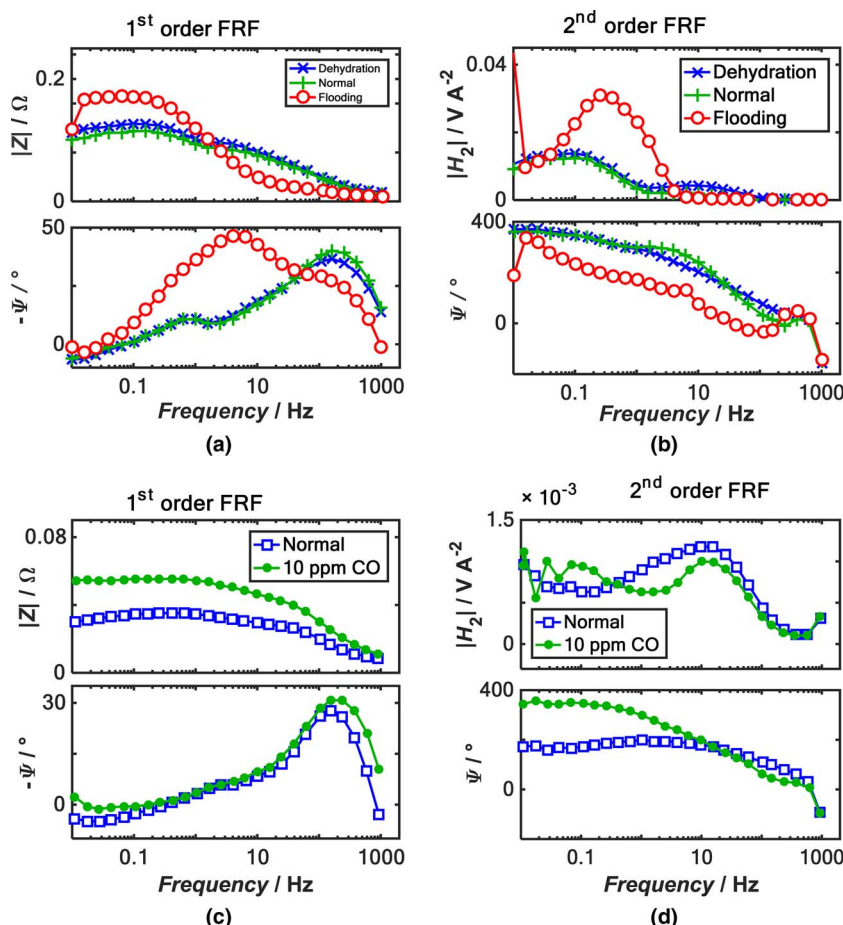


Figure 9. (a) 1st order FRF measured at $i_{dc} = 0.231 \text{ A cm}^{-2}$ and $i_{ac0} = 58 \text{ mA cm}^{-2}$ under dehydrated, normal and flooding conditions (b) 2nd order FRF measured at $i_{dc} = 0.231 \text{ A cm}^{-2}$ and $i_{ac0} = 58 \text{ mA cm}^{-2}$ under dehydrated, normal and flooding conditions (c) 1st order FRF measured at $i_{dc} = 0.231 \text{ A cm}^{-2}$ and $i_{ac0} = 96 \text{ mA cm}^{-2}$ under normal conditions and anodic poisoning with 10 ppm CO (d) 2nd order FRF measured at $i_{dc} = 0.231 \text{ A cm}^{-2}$ and $i_{ac0} = 96 \text{ mA cm}^{-2}$ under normal conditions and anodic poisoning with 10 ppm CO. Adapted from Ref. 6.

of 2nd FRF of methanol oxidation kinetics can provide additional information to distinguish between candidate mechanisms.

Total Harmonic Distortion (THD) is another method used to characterize the nonlinear behavior of electrochemical systems.^{61,62} Under pseudo-potentiostatic conditions, THD can be defined as the ratio of the sum of amplitudes of responses at all higher harmonics to that of the fundamental frequency.

$$THD = \frac{\sqrt{\sum_{k=2}^{\infty} |i_k|^2}}{i_1} \quad [28]$$

where i_k is the magnitude of the current at k^{th} harmonic. Under pseudo-galvanostatic conditions, the corresponding THD could be defined using the potential results. THD does not use phase data since phase values of different harmonics cannot be meaningfully combined. The nonlinear behavior of DMFC was investigated using THD analysis.^{25,26} THD and EIS data were recorded using Zahner impedance measurement units (IM6e and PP201) at 2.5 A dc current and using 0.5 A input signal amplitude. Frequency of the input signal was scanned from 5 kHz to 0.01 Hz. The THD spectra and analysis were performed both in half cell and full cell modes. Potential responses from the first harmonic to tenth harmonic were obtained and the experimental THD spectra with different methanol concentrations are shown in the Fig. 11.

THD increases with concentration in the frequency range from 0.1 Hz – 0.63 Hz and decreases with concentration from 0.01 Hz – 0.04 Hz frequency. Thus, at certain frequency intervals, the experimental THD spectra exhibited clear correlation with methanol concentration in the anodic feed. Hence it was proposed that THD analysis can be used to detect the methanol concentration during DMFC operation. The variation of THD spectra with methanol concentration and the

methanol oxidation mechanism in DMFC were also investigated.²⁵ The observed features of the THD spectra were reproduced qualitatively by assuming a three step mechanism for methanol oxidation with Kauranen-Frumkin/Temkin kinetics rather than a one-step mechanism. THD values were computed by transforming the potential results to frequency domain using Fourier transform and resolving into fundamental and higher harmonic components. Experiments conducted in the full cell mode of DMFC operation also showed similar results, indicating that THD analysis can be employed for sensing methanol concentration during fuel cell operation. Thus THD can help in identifying the reaction mechanism as well as in identifying the changes in operating conditions.

Other Applications.—The electrochemical properties of industrial lubricants at different stages of degradation were characterized using higher harmonic NLEIS technique.⁶³ Lubrication samples were subjected to oxidation engine test. Samples after 20 h, 40 h and 80 h of test runs, as well as fresh samples (0 h) were analyzed using a two electrode setup. Current data was acquired in the frequency range of 20 MHz to 5 mHz at an amplitude (E_{ac0}) of 1.5 V. Since there was no water in the samples, such large amplitudes could be employed. Fundamental, second and third harmonics were recorded and analyzed to investigate the processes affecting the low frequency impedance behavior of the lubricant samples. The reaction involved in the adsorption process was described by Butler Volmer kinetics and the diffusion impedance was described by finite transmission boundary diffusion model. The experimental linear impedance did not show a monotonic trend with engine test duration and hence cannot be used to ascertain the extent of oil degradation. On the other hand, 2nd harmonic data showed a clear monotonic trend with test duration. Physical model in combination with EEC, representing the charge and mass transfer processes and adsorption processes at the electrode electrolyte

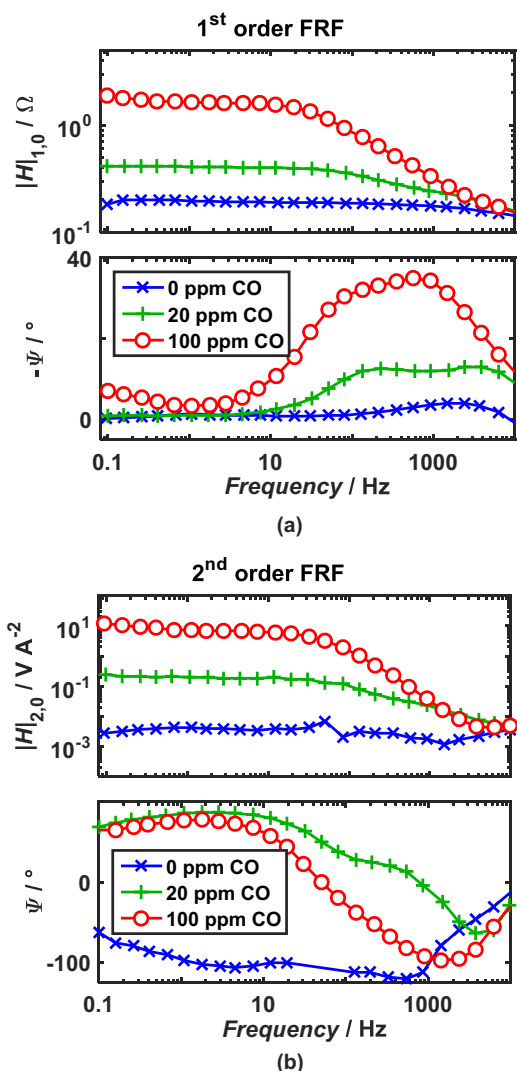


Figure 10. Measured NFRA spectra in H_2/H_2 operation with various anode CO content, at $i_{dc} = 100 \text{ mA cm}^{-2}$. The perturbation amplitude $i_{ac0} = 90 \text{ mA cm}^{-2}$ for no CO poisoning and $i_{ac0} = 60 \text{ mA cm}^{-2}$ for 20 and 100 ppm CO poisoning. (a) Magnitude and phase of 1st order FRF, $H_{1,0}$ (b) Magnitude and phase of 2nd order FRF, $H_{2,0}$. Adapted from Ref. 7.

interface was developed to explain the low frequency behavior. The model and the kinetic parameters which result in best fit of the experimental fundamental and higher harmonic data was obtained using numerical minimizations routines. The best fit EEC model is shown in Fig. 12a.

Studies on microparticle biomarker samples derived from monocytes (white blood cells) suggested that potential-induced microparticle lysis and subsequent redox reactions between the electro active components may be the reason for the low frequency impedance of these samples.⁶⁴ This was further analyzed using NLEIS technique.⁴⁰ Nonlinear EIS measurements were conducted using an Impedance analyzer (Novocontrol GmbH, Hundsangen Germany) with ZG4 cell adapter. A three electrode system configuration with gold plated copper working and counter electrodes and Ag/AgCl reference electrode was employed. Fundamental and 2nd harmonic impedance of microparticle biomarker samples recorded at different dc potentials were fitted to a process model shown as equivalent circuit in Fig. 12b. The model accounts for adsorption of microparticles, double layer as constant phase element, diffusion and simple electron transfer reaction of lysed components. It was assumed that the concentration of redox couple could be described in Nernst equation. Mass transport

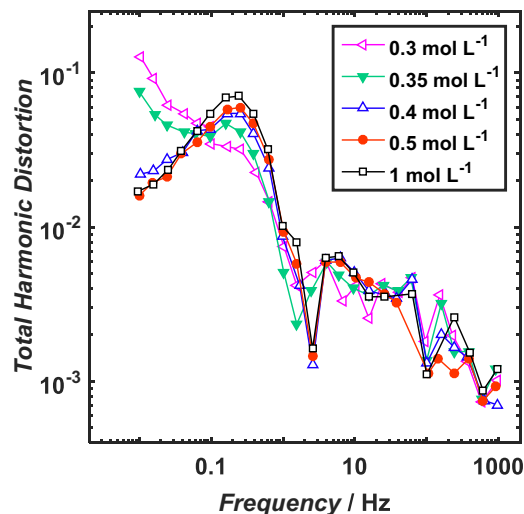


Figure 11. Experimental THD spectra of the DMFC in half cell mode for different methanol concentrations at $i_{dc} = 96 \text{ mA cm}^{-2}$, $i_{ac0} = 19.2 \text{ mA cm}^{-2}$. Adapted from Ref. 25.

process was described by finite absorbing boundary diffusion and the microparticle adsorption process was assumed to occur via a single step mechanism. The nonlinear terms in the governing equations were expanded in power series and were separated into harmonic components. Second harmonic impedance was defined as $\overline{E_{2\omega}} / \overline{i_{2\omega}}$ where $\overline{E_{2\omega}} = (E_{ac0})^2 \sin(\omega t)$. The physical parameters were obtained by fitting the experimental data. NLEIS results indicated that the applied potential caused lysis of the microparticle membranes. However, quantification of microparticles by NLEIS was not recommended and for that purpose, linear EIS at various dc potential was suggested instead.

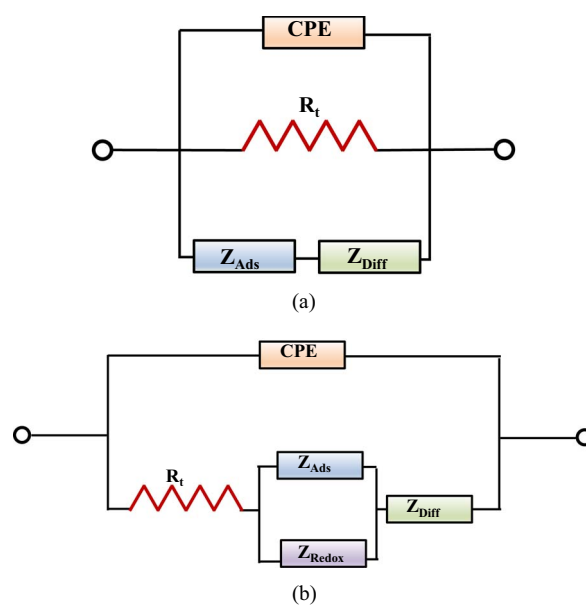


Figure 12. Equivalent circuit model for the low frequency impedance segment. Z_{Diff} – Diffusion impedance, Z_{Ads} – Adsorption impedance and Z_{Redox} – microparticle lysed component redox process. (a) EEC for lubricants and (b) for biomarker samples obtained from monocytes. Adapted from Refs. 63 and 40 respectively.

Prospectus

Although experimental and theoretical results on NLEIS have been published since 1990s, the reports on application of NLEIS to investigate practical problems have been rather limited, while the linear EIS has found extensive usage in a variety of fields. EIS is a standard feature in most of the commercial electrochemical instruments and usually it comes with a convenient user interface, facilitating easy acquisition of linear EIS data. In contrast, to acquire NLEIS of any system, researchers often have to assemble multiple instruments possibly from different manufacturers and write control, data acquisition and digital signal processing (DSP) programs. For many users, this is a significant road block toward performing NLEIS experiments. The duration of experiment and signal to noise (S/N) ratio are two important metrics of any measurement technique. During NLEIS measurement using Solartron 1250 FRA, the higher harmonic data recorded showed poor S/N ratio at frequencies above 3 kHz.⁶⁵ This is presumably because traditional EIS is more widely used and the data acquisition and DSP electronics may be optimized to achieve good S/N ratio mainly at fundamental frequency. Similarly, when PSM 1735 PSA was employed, noise control on the input signal was poor at frequencies above 10 kHz.⁶⁵ More importantly, duration of a typical NLEIS measurement is significantly longer than that of traditional EIS measurement. When an ADC was used to record NLEIS data, 100 cycles were needed to achieve a good S/N ratio.⁶⁵ The lowest frequency at which data was recorded was 50 mHz and the corresponding NLEIS acquisition time would be 2000 s for this single datum. In another study, 20 cycles were employed at each frequency and the lowest frequency employed was 100 mHz, with a corresponding data acquisition time of 200 s for this datum alone. In contrast, traditional EIS measurement in the entire frequency range of 100 kHz to 100 mHz can be completed in less than 200 s. It is highly desirable to evolve a faster NLEIS data acquisition methodology while maintaining a good S/N ratio.

In case of EIS, oscilloscopes and lock-in amplifiers were used in the early days, but now FRAs are universally employed. There is no standardization of experimental method to acquire NLEIS data yet: phase sensitive detectors, or FRA or ADC with further processing of data by the user, have all been employed in the recently published results.^{8,9,65,66} Yet another difficulty faced by the NLEIS user is the lack of fast analysis using EEC and the limited number of studies available in the literature.⁶⁵ There are several reports which present traditional EIS data and analysis using EEC and to a lesser extent using RMA, which enable users to compare and interpret their experimental data. There are fewer publications on NLEIS which can be used as reference points for comparison and interpretation, but this also points to many opportunities available for investigation. Admittedly, the analysis of NLEIS data is more challenging than that of linear EIS and lack of easy to use analysis software is an impediment. We believe that development of uncomplicated hardware and software and standardization of analysis methods will enable wider application of NLEIS. Only recently, reports demonstrating actual utility of NLEIS have been published. We anticipate that in the near future, a number of empirical and semi-empirical studies on NLEIS application would be reported, and that at later stages, with accumulation of large NLEIS data, rigorous NLEIS analysis providing better insight into electrochemical systems would become available.

Conclusions

EIS is a versatile technique used to probe electrochemical interfaces, and it is used in a variety of applications such as corrosion, fuel cell, biosensor etc. Traditional (linear) EIS is powerful and in many applications, it provides sufficient information to characterize the system. NLEIS can be thought of as an extension of EIS, and it can give additional information compared to EIS. In NLEIS, the response at fundamental frequency to large amplitude perturbations can yield some additional information, and recording the response at higher harmonics can yield significantly more information, since contributions from double layer capacitance to higher harmonics are

usually small. For a simple redox reaction, extensive analysis has been performed and experimental results have been compared. For electrochemical reactions in non-blocking electrodes, expressions for the low frequency limit of the impedance at fundamental frequency ($R_{p,NL}$) have been developed under potentiostatic and galvanostatic conditions. In case of complex reactions, there is limited information available in literature, especially at higher harmonics.

To predict the NLEIS response of a system, Taylor series expansion or Fourier expansion or numerical methods can be used. Other variables such as primary response coefficient or Volterra kernel have been extracted from the higher harmonics, to remove the dependency on applied perturbation, and analyzed. THD can be used as a measure of nonlinearity and employed to draw conclusions based on empirical understanding, but rigorous analysis and interpretations would be challenging since phase information is discarded in THD. Recent reports illustrate that NLEIS can identify the physical process unambiguously in certain cases where linear EIS results are indistinguishable. It is likely that in the near future, NLEIS will be employed for semi-empirical understanding of electrochemical systems, and with further development of rigorous analysis, it would become a more common technique offering additional insights into the physico chemical processes occurring at the electrochemical interfaces.

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